

Swimming with Data: The Cloud Revolution in Salmon Farming



Who are we?



Kasper Kalfas

Cloud Architect at Software Mind

Turns raw data into cloud business solutions.



Daniel Pisarek

AI & Cloud Architect at Software Mind

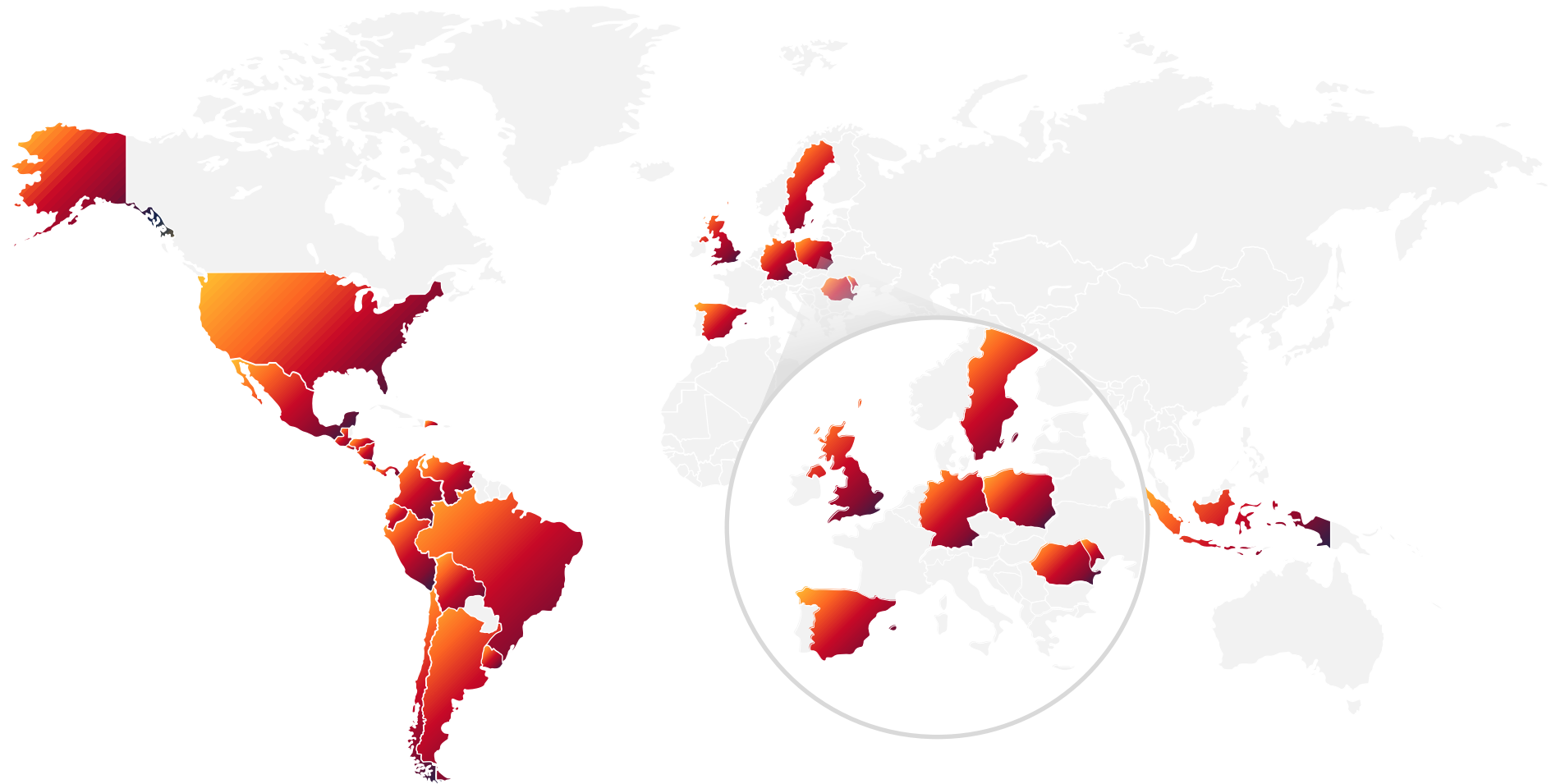
Blends AI with cloud solutions for innovation.

About Software Mind

Software Mind, a global technology partner, engineers software that creates impact by delivering scalable, cross-functional teams who manage software life cycles and drive innovation. Experience with cloud, data science, AI and other leading tech enables our experts to accelerate digital transformations.

Our expertise includes:

- Cloud architecture
- Data engineering
- AI & ML
- Software development

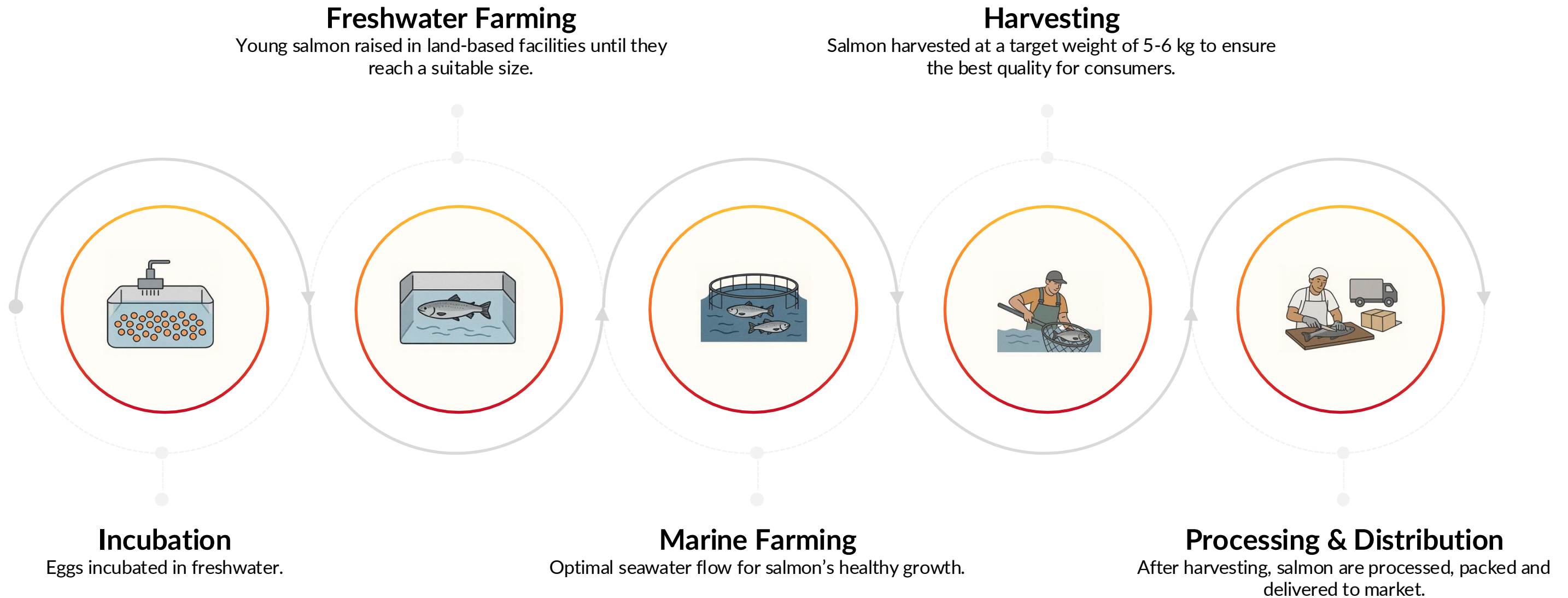


Client success story: Grieg Seafood

- Grieg Seafood (GSF) has decades of aquaculture experience in Norway and Canada.
- They produce over 100,000 tons of salmon annually.
- A traditional salmon company that uses technology to drive innovation.



The lifecycle of salmon farming



This is what a salmon farm looks like



The challenges that we met

1

A traditional salmon company that wants to be innovative.

2

GSF invested in **electronics sensors (IoT)** that collected data from farms.

3

But data was stored in **multiple databases** located in **Norway and Canada**.

6

Data collection created bottlenecks, which slowed down analysis and decision-making.

5

There was **no centralized data platform**.

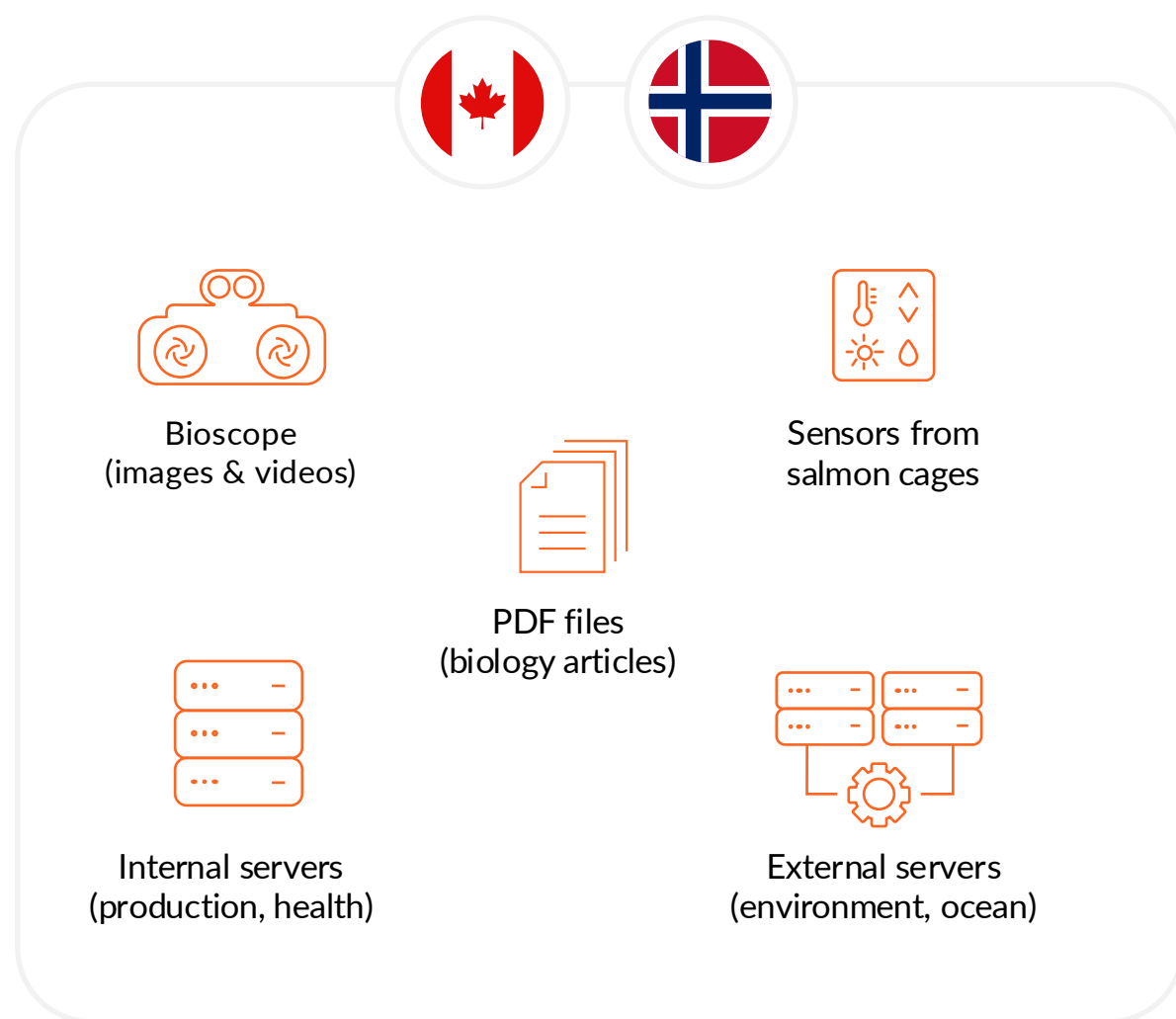
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Systems were **not integrated** – they couldn't "communicate" with each other.

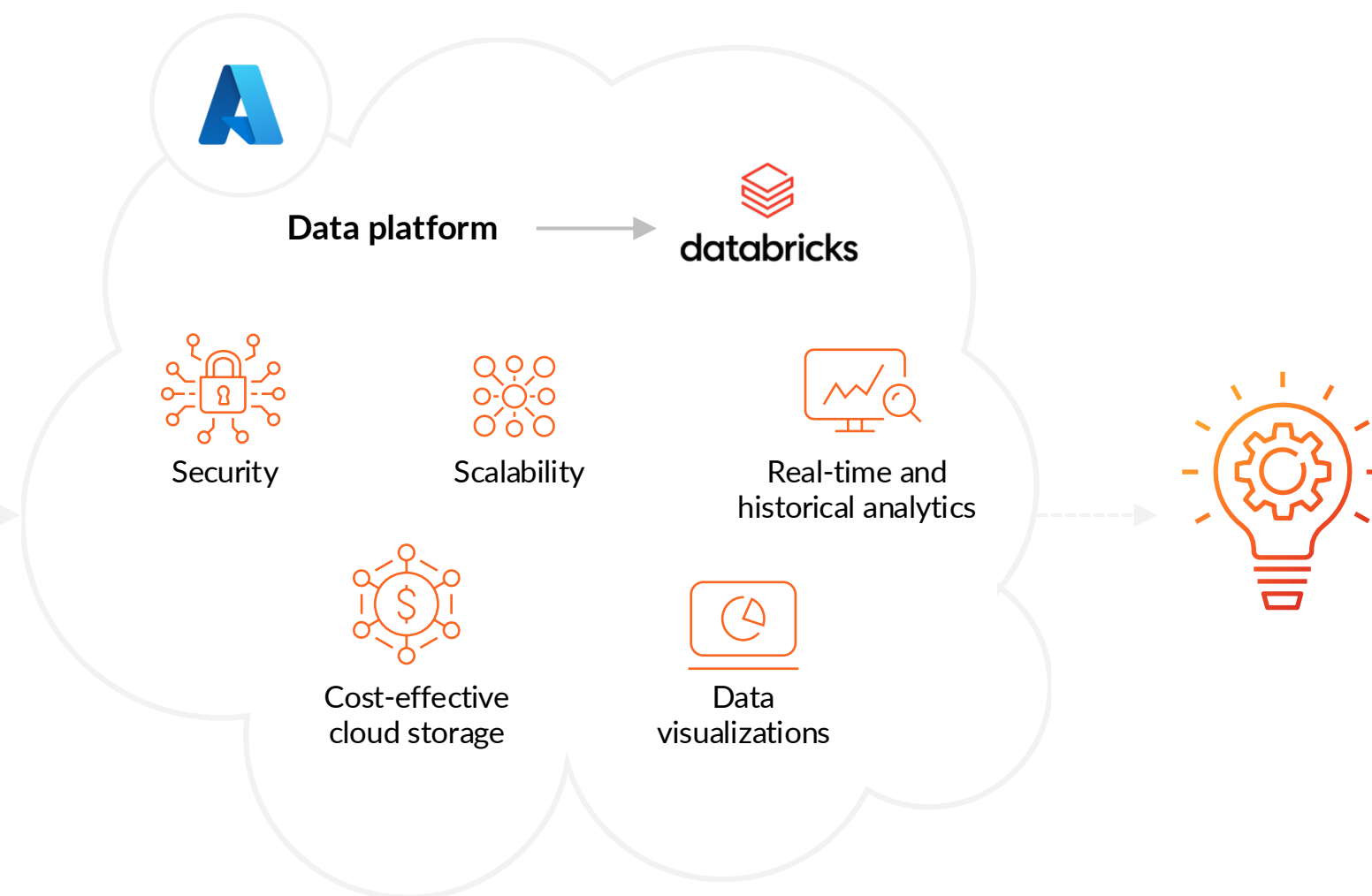


From data to insights

Data Sources



Cloud data processing



 New ideas were born...



Data against nature's challenges



Sea lice cause stress
and open wounds



Jellyfish stings damage gills
and sea plankton reduce oxygen

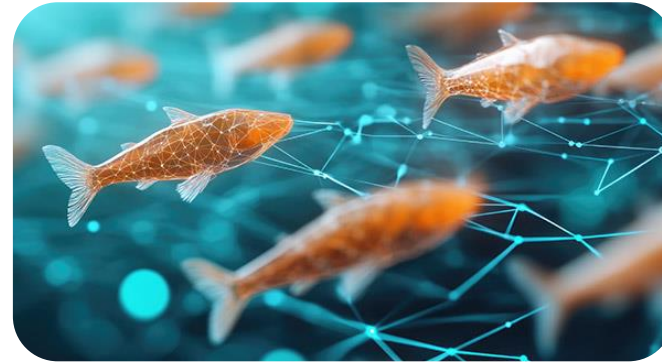
Using our cloud data platform, GSF can now anticipate and mitigate risks.
This ensures sustainable aquaculture operations despite natural challenges.

Empowering salmon farming with AI-Agent (SalmoPedia)



CHALLENGE:

After systematizing and organizing our data, we faced the challenge of making it accessible and actionable for salmon farmers



SOLUTION:

Implementation of a specialized AI agent that analyzes farming data, scientific articles and biological reports



BENEFITS:

- Simplified access to specialized knowledge
- Proactive data gathering through intelligent follow-up questions
- Enhanced decision-making through contextual knowledge sharing

Salmon harvest



CHALLENGE:

Farmers used to rely on intuition to decide the right harvest time



SOLUTION:

but now, thanks to real-time data from sensors and underwater photos



BENEFITS:

they can harvest at the perfect moment — improving fish quality and reducing waste.

What have we achieved?



Global Knowledge Integration

- Eliminated information silos between Norwegian and Canadian operations



Enhanced Operational Excellence

- Improved farm productivity through shared best practices
- Reduced production risks with cross-regional insights
- Optimized harvesting decisions with predictive intelligence

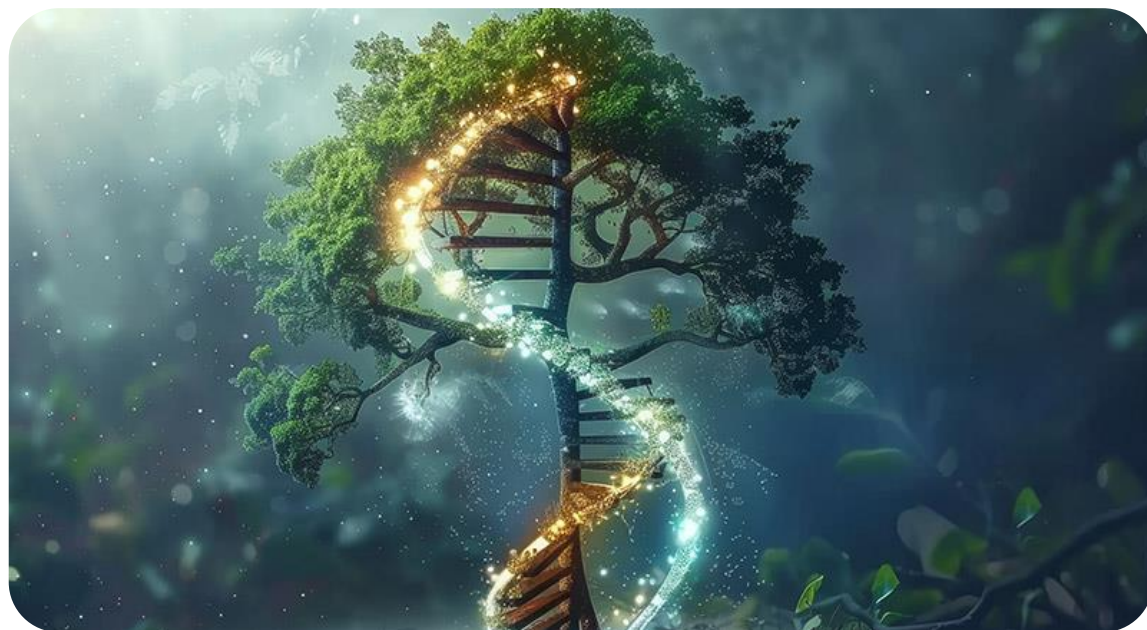


Business Transformation

- Increased innovation with AI-powered decision-making support
- Streamlined operations through centralized knowledge management



Future innovations



Tracing Ancestry in graph database.

Track specific salmon groups across multiple generations in graph database.



Monitoring Salmon Feed Quality

Feed directly affects fish growth, health and product quality.
Monitor feed composition and response during farming.

Conclusion

Questions?

What is hidden in your isolated data?

Is your digital foundation limiting your future?

Are you seeing patterns or just collecting numbers?

What becomes possible when AI amplifies both team knowledge and business processes?

What would you do differently if you could see the future?

Key insights

Connecting data domains and sources: Unifying separate knowledge systems creates collective intelligence.

Cloud as a foundation: Centralized infrastructure enables secure integration and scalability that powers innovation.

Strategic data processing: Converting raw information into actionable insights directly impacts business outcomes.

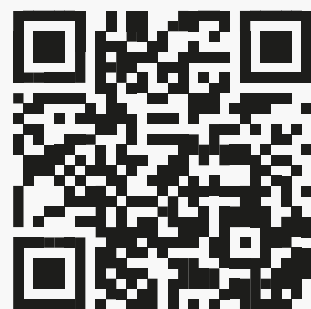
AI-enhanced expertise: Intelligent systems that continuously learn create compounding value beyond human limitations.

Predictive business models: Combining diverse data sources enables forecasting that transforms strategic planning.

Contact



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**What were the salmon doing before
we implemented data algorithms?**

They were just swimming in data.